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Steganographic Channels as a Countermeasure to Prompt Injection and Man-in-the-Middle Manipulation in Agentic Pipelines

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Abstract:

The rapid development of agentic artificial intelligence has transformed AI systems from passive response-generating applications into systems capable of planning tasks, interacting with external tools, retrieving information, maintaining memory, and taking actions with limited human intervention. While these capabilities increase efficiency and autonomy, they also introduce cybersecurity challenges. Information exchanged between agents, tools, databases, APIs, documents, and intermediary components may be manipulated before reaching its intended destination. Prompt injection, indirect prompt injection, message alteration, compromised intermediaries, and unauthorized tool use can affect the reliability and security of agentic workflows. This paper examines a security approach combining image-based Least Significant Bit (LSB) steganography with HMAC-SHA256 authentication to provide an additional layer of message concealment and integrity verification. The proposed prototype conceals a message and authentication information within a carrier image and subsequently extracts and verifies the message. A controlled Man-in-the-Middle (MITM) simulation tests whether unauthorized message modification can be identified. The study follows an applied experimental methodology involving system design, implementation, controlled testing, and security analysis. Findings indicate that the prototype successfully retrieves concealed messages and identifies alterations when modified content produces an authentication value different from the original. However, authentication cannot determine whether an authorized message is malicious. Secure key management, authorization, access control, replay protection, prompt-injection detection, monitoring, and human approval for high-risk actions are therefore necessary for a comprehensive security architecture. The study concludes that combining concealment with cryptographic integrity checking provides a useful supplementary security mechanism for agentic AI communication.

Keywords: Agentic Artificial Intelligence, Prompt Injection, Steganography, HMAC-SHA256, Message Integrity, Man-in-the-Middle Attack, AI Security

1. Introduction

Artificial intelligence has progressed from systems designed primarily to generate predictions or responses to increasingly autonomous systems that can plan activities, access external information, use software tools, coordinate multiple operations, and make decisions based on changing environmental information. Such systems are commonly described as agentic AI systems. Unlike a conventional chatbot, an agentic system may interact with websites, databases, files, APIs, applications, and other AI agents. The uploaded project identifies an agentic pipeline as a system in which an AI agent can understand a task, select an appropriate course of action, use tools, obtain information, and use that information for subsequent decisions.

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The increasing autonomy of these systems also expands their attack surface. An agent may simultaneously process trusted instructions and information obtained from sources that cannot automatically be considered trustworthy. A malicious instruction hidden inside a webpage, document or email can potentially be interpreted as an instruction rather than ordinary data. This creates the problem of prompt injection, particularly indirect prompt injection, where malicious content is embedded in an external source that an AI system subsequently retrieves. Research by Greshake et al. (2023) demonstrated that indirect prompt injection can exploit the blurred boundary between instructions and data in LLM-integrated applications.

The problem becomes more serious when an AI agent possesses tools capable of taking consequential actions. A manipulated message may therefore have effects extending beyond the generation of an inaccurate answer. It may influence another agent, trigger an API call, modify a file, expose information, or initiate another operation. Current OWASP guidance identifies prompt injection, excessive agency, supply-chain vulnerabilities, sensitive information disclosure and other risks as important security concerns for AI applications.

Another security concern arises when information travels through several components before reaching the final AI agent. An intermediary may modify, replace, filter, summarize, or otherwise transform information. In such circumstances, the receiving agent may know the identity of the apparent sender but still have difficulty determining whether the message is identical to the original. The uploaded project specifically identifies this as a concern within agentic pipelines and proposes message integrity verification as an additional security layer.

The paper explores a prototype approach that combines Least Significant Bit steganography and HMAC-SHA256 authentication. Steganography is used to conceal the message within an apparently ordinary image, while HMAC-SHA256 is used to verify whether the recovered message has been modified. The approach is tested through a controlled MITM attack simulation in which the original message is intentionally changed while its original authentication information is retained.

The central premise is that message concealment and message authentication address different security requirements. Steganography can reduce the visibility of the communication, but it does not establish authenticity. HMAC can establish whether the authenticated content has changed, but it does not conceal the content. Combining the two mechanisms therefore provides complementary protection.

2. Literature Review

2.1 Agentic Artificial Intelligence and Emerging Security Risks

Agentic AI represents an important development in the evolution of AI applications because the system can move beyond producing information and can participate in multi-step workflows. Agents may plan activities, call tools, retrieve information, interact with external systems, and pass information to other components. Consequently, security must consider not only the model itself but also the complete environment surrounding the model.

The security implications of this development are increasingly recognized by professional cybersecurity organisations. OWASP's work on LLM and generative AI security identifies prompt injection, insecure output handling, supply-chain vulnerabilities, sensitive information disclosure, excessive agency and related risks. More recent OWASP guidance specifically addresses agentic applications and highlights risks associated with goal hijacking, tool misuse, identity and privilege abuse, supply-chain vulnerabilities, memory and context poisoning, and other agent-specific threats.

The literature therefore indicates that conventional cybersecurity controls cannot simply be transferred to agentic AI without modification. Security mechanisms must account for the fact that natural-language information can influence automated decision-making.

2.2 Prompt Injection

Prompt injection occurs when malicious content influences an AI system's interpretation or behaviour. Direct prompt injection involves an attacker intentionally supplying malicious instructions to the model, whereas indirect prompt injection occurs when malicious instructions are embedded in external content subsequently processed by the system.

Greshake et al. (2023) demonstrated that indirect prompt injection can be particularly problematic because the attacker may not need direct access to the AI interface. Malicious content can be placed within information that an AI

application later retrieves. Their research showed potential impacts including manipulation of application functionality, data theft and unauthorized interaction with APIs.

The paper similarly recognizes that agentic systems may process instructions and external information together, creating difficulty in distinguishing trusted commands from untrusted content. Current OWASP guidance recommends external-content segregation, input and output filtering, least-privilege access, adversarial testing and human approval for high-risk operations.

2.3 Message Authentication

Message authentication is an established cybersecurity concept intended to allow a recipient to determine whether a message has been altered and whether it was generated using the appropriate secret key. Krawczyk, Bellare, and Canetti (1997) introduced HMAC as a mechanism based on cryptographic hash functions and a shared secret key.

HMAC-SHA256 is particularly relevant to the present study because SHA-256 produces a fixed-length message digest, while HMAC incorporates a secret key into the authentication process. NIST's Secure Hash Standard recognizes SHA-256 as a secure hashing algorithm used to generate message digests and detect changes to messages.

The principle underlying the proposed system is straightforward. The sender calculates an authentication value for the original message. At the receiving end, the authentication value is recalculated using the recovered message and the same secret key. If the values match, the message is considered unchanged under the assumed key and algorithm. If they differ, the system identifies the message as altered.

2.4 Steganography

Steganography differs from encryption because its principal purpose is to conceal the existence of information rather than merely transform the information into an unreadable form. Digital steganography can embed information within images, audio, video or text.

The present paper employs LSB image steganography. In this technique, information is embedded by making small changes to the least significant bits of image-pixel values. Because the changes are generally small, the resulting image can appear visually similar to the original carrier. The uploaded project uses this method to conceal a message while retaining the ordinary appearance of the carrier image.

Steganography nevertheless has limitations. Image compression, resizing, format conversion or other transformations can interfere with embedded information. Consequently, steganography should not be treated as an independent guarantee of communication security.

3. Research Methodology

The research follows an applied experimental and prototype-based methodology. The purpose was not to develop a theoretical cryptographic algorithm but to construct and evaluate a working demonstration of message concealment and integrity verification.

The research process consisted of five major stages.

Stage 1: Identification of the Security Problem

The paper first examined communication within agentic pipelines and identified message alteration, prompt injection and trust between interconnected components as major concerns.

Stage 2: System Design

A prototype architecture was designed around three major functions:

- message hiding;
- message extraction and authentication; and
- MITM attack simulation.

The paper identifies Python and Streamlit as the implementation technologies and describes these three modules as the central components of the application.

Stage 3: Message Protection

A user enters a message and selects an image as a carrier. The system generates authentication information using the message and a secret key. The message and associated information are subsequently embedded within the carrier image.

Stage 4: Message Recovery and Verification

At the receiving end, the protected image is uploaded. The concealed message is extracted and the authentication value is recalculated. The newly calculated value is compared with the original authentication information.

Stage 5: Controlled MITM Simulation

The original message is intentionally modified while the original authentication value is retained. The system recalculates the authentication value of the modified message. Because the content has changed, the resulting value does not match the original value, causing verification to fail. The paper records successful extraction, authentication of unchanged messages and failure of authentication after simulated modification.

This methodology provides a controlled environment for examining the security concept without conducting an actual attack against an external network or third-party system.

4. Research Gap

The literature indicates substantial work on prompt injection, cryptographic authentication and information hiding. However, the security requirements of agentic AI create a need for mechanisms that can operate at the communication layer between different components of an AI workflow.

The specific contribution examined in this study is therefore not the invention of a new cryptographic algorithm or steganographic technique. Instead, it lies in demonstrating how established techniques can be combined into a simple prototype for agent-to-agent communication and tamper detection. The research also explicitly examines the boundary of such an approach: successful authentication confirms message integrity under the authentication mechanism, but it does not establish that the original message is safe or that the sender is authorized to perform every possible action.

5. Research Objectives

The research paper was undertaken with the following objectives:

- To examine major security risks associated with communication within agentic AI pipelines.
- To understand the role of prompt injection and intermediary message manipulation in agentic AI environments.
- To design a prototype that combines steganography with cryptographic message authentication.
- To implement HMAC-SHA256 for message integrity verification.
- To examine whether unauthorized modification of an authenticated message can be detected.
- To simulate a controlled MITM manipulation scenario.
- To evaluate the usefulness and limitations of the proposed security mechanism.
- To suggest additional controls required for practical deployment in agentic AI environments.

6. Proposed Security Architecture

The proposed architecture consists of a sender, carrier medium, communication process and receiving component.

At the sender side, the message is first prepared. An HMAC-SHA256 value is generated using a secret key. The message and relevant authentication information are then embedded into an image using LSB steganography.

The resulting image functions as the protected carrier. At the receiving side, the system extracts the concealed message and obtains the authentication information. It then independently calculates the HMAC for the extracted message.

Two outcomes are possible:

Authenticated message:

Original authentication value = Recalculated authentication value

In this situation, the message is treated as unchanged under the assumed security conditions.

Tampered message:

Original authentication value ≠ Recalculated authentication value

In this situation, the system identifies a modification and prevents the altered communication from being treated as the original authenticated message.

The approach is consistent with the established purpose of HMAC, which is to provide a means of verifying message integrity using a shared secret key.

The architecture also incorporates a controlled MITM simulation. Rather than attempting to reproduce every technical characteristic of a real network attack, the prototype changes the message after authentication and observes whether the verification mechanism detects the change.

7. Current Scenario of Agentic AI Security

The security environment surrounding generative and agentic AI is changing rapidly. AI systems are increasingly being connected to enterprise information, applications, databases, APIs and productivity tools. As autonomy increases, the consequences of security failures can also increase.

The current security discussion has moved beyond the traditional concern of incorrect AI output. Security researchers and standards organisations increasingly examine the possibility that AI agents may be manipulated into taking inappropriate actions.

OWASP's current agentic-AI security work identifies agent goal hijacking, tool misuse, identity and privilege abuse, supply-chain vulnerabilities, unexpected code execution, and memory or context poisoning among the emerging risks.

NIST's AI Risk Management Framework and its Generative AI Profile similarly emphasize structured risk management throughout the AI lifecycle. The NIST Generative AI Profile provides organizations with guidance for identifying and managing risks associated with generative AI.

These developments indicate that security should be integrated into the design of AI systems rather than added after deployment.

For agentic systems, communication integrity is particularly important because information may pass between multiple components. A message that is changed between the originating agent and receiving agent may influence subsequent reasoning. The uploaded project therefore places emphasis on verifying information before it is used by another component.

8. Major Issues and Challenges

8.1 Prompt Injection

Prompt injection remains a fundamental challenge because an AI model may not reliably distinguish between trusted instructions and untrusted text. External documents, websites and emails may contain content that attempts to influence the agent's behaviour.

8.2 Indirect Prompt Injection

The danger increases when an agent automatically retrieves external information. An attacker can potentially place malicious instructions in a source that the agent later reads. The agent may consequently treat hostile content as an instruction.

8.3 Message Manipulation

Information can be altered while moving through intermediaries. A receiving agent therefore needs mechanisms to determine whether the received content corresponds to the original authenticated content.

8.4 Excessive Agency

If an AI system has more privileges than necessary, a manipulated message may have disproportionately serious consequences. Least-privilege design is therefore important.

8.5 Key Management

The prototype uses a predefined authentication arrangement. A production system would require secure key generation, storage, rotation and distribution. If an attacker obtains the secret key, the attacker may be able to generate valid authentication values for modified messages.

8.6 Replay Attacks

Authentication alone does not necessarily determine whether a valid message is new. An attacker might reuse an earlier legitimate message. Timestamps, nonces, sequence numbers or unique message identifiers can provide additional protection.

8.7 Malicious but Authentic Messages

This is one of the most important limitations of the proposed approach. Authentication answers the question: **"**Has the authenticated message changed?" It does not necessarily answer: "Is the message safe?"

An authorized source could generate a dangerous instruction. If the message remains unchanged, the HMAC would still verify successfully. The uploaded project explicitly recognizes that authentication cannot independently detect malicious instructions within an otherwise authentic message.

8.8 Steganographic Robustness

Image-based LSB steganography may be affected by compression, resizing and file-format conversion. Consequently, the technique may not be suitable for every communication environment.

8.9 Lack of Complete Authorization

Authentication of a message does not automatically mean that the sender has permission to request every operation. Authentication and authorization should therefore remain separate controls.

9. Findings

9.1 Message Concealment was Achievable

The prototype successfully embedded messages within carrier images and subsequently extracted the concealed information. The Hide Message and Extract Message modules therefore demonstrated the basic feasibility of the proposed workflow.

9.2 HMAC-SHA256 Detected Message Modification

When the original message was retained, the recalculated authentication value matched the original authentication value. When the message was deliberately modified, the values no longer matched. This allowed the system to identify the message as altered.

9.3 Controlled MITM Simulation Demonstrated Tamper Detection

The MITM demonstration successfully showed that changing an authenticated message without possessing valid authentication information causes verification failure. The system consequently distinguishes between an unmodified message and a manipulated message.

9.4 Concealment and Authentication Serve Different Purposes

The research confirms that steganography and authentication should not be considered interchangeable. Steganography hides information, whereas HMAC provides integrity and authentication. Their combination can therefore provide complementary security functions.

9.5 The Proposed Approach is Supplementary Rather than Comprehensive

The prototype does not provide complete protection against prompt injection, malicious authorized instructions, compromised keys, replay attacks, excessive privileges or all possible network attacks. The project itself characterizes the implementation as a prototype requiring additional controls for practical deployment.

10. Suggestions and Recommendations

10.1 Implement Secure Key Management

Future implementations should replace fixed or predetermined authentication arrangements with secure key-generation, storage, rotation and distribution mechanisms. Hardware-backed key storage or enterprise key-management systems may be considered for high-value deployments.

10.2 Adopt Least-Privilege Access

AI agents should receive only the permissions necessary for their assigned tasks. Tool access should be restricted so that a manipulated instruction cannot automatically produce unrestricted consequences.

10.3 Introduce Prompt-Injection Detection

Message authentication should be supplemented with mechanisms capable of identifying suspicious instructions, conflicting commands and potentially malicious external content. OWASP recommends input/output filtering, external-content segregation, adversarial testing and human approval for high-risk operations.

10.4 Introduce Replay Protection

Timestamps, unique message identifiers, nonces or sequence numbers should be incorporated so that an old authenticated message cannot simply be reused.

10.5 Separate Authentication from Authorization

A valid HMAC should demonstrate message integrity under the relevant key, but it should not automatically grant permission to perform an action. Separate authorization policies should determine whether the requested operation is allowed.

10.6 Maintain Audit Logs

Authentication failures, rejected messages, unusual agent behaviour and high-risk tool calls should be recorded. Such logs can support security investigations and continuous monitoring.

10.7 Use Human Approval for High-Risk Operations

Financial transactions, deletion of important data, privilege changes, external communication and other consequential actions should require human approval or an additional authorization layer.

10.8 Improve Steganographic Robustness

Future research should examine more resilient steganographic approaches and alternative carrier formats. The impact of compression, resizing, transformation and transmission through different platforms should be experimentally evaluated.

10.9 Conduct Adversarial Testing

The prototype should be expanded into a larger testing environment involving prompt injection, indirect prompt injection, replay, key compromise, malicious external documents and manipulated tool outputs.

11. Limitations of the Study

- First, the prototype operates within a controlled environment and therefore does not represent the complete complexity of a production-scale agentic AI ecosystem.
- Second, the MITM attack is simulated rather than conducted against an actual distributed communication network. Consequently, the experiment demonstrates the effect of message manipulation but does not evaluate every technique that could be employed by a real attacker.

- Third, the prototype does not implement comprehensive key-management infrastructure. This limits its direct applicability to enterprise systems.
- Fourth, LSB steganography may be affected by transformations to the carrier image. Therefore, the method requires further evaluation under realistic transmission conditions.
- Finally, HMAC verifies integrity but does not establish that the original content is safe. A legitimate but malicious instruction could pass the authentication check. The project therefore correctly identifies prompt-injection detection, authorization, access control and monitoring as complementary requirements.

12. Future Scope

- First, a multi-agent environment could be developed in which several AI agents exchange authenticated messages. This would permit examination of integrity verification at multiple points within an agentic pipeline.
- Second, digital signatures could be investigated as an alternative or complement to shared-key authentication where independent verification of message origin is required.
- Third, authenticated encryption could be incorporated to provide confidentiality and integrity together.
- Fourth, the system could be extended beyond images to audio, video and document-based carriers. Comparative experiments could evaluate robustness, capacity, computational overhead and detectability.
- Fifth, the prototype could incorporate automated prompt-injection detection. This would enable a two-stage security model in which a message is first authenticated and then examined for potentially harmful or unauthorized instructions.
- Sixth, the system could incorporate timestamps, message identifiers and session information to provide replay protection.
- Finally, future research should investigate integration with AI security frameworks and agent-control mechanisms. The recent OWASP focus on agentic applications demonstrates that security must increasingly address not only the model but also identity, privileges, tools, memory, supply chains and autonomous actions.

13. Conclusion

The emergence of agentic AI introduces a new dimension to cybersecurity because AI systems are increasingly capable of interacting with external information and taking actions with limited human intervention. In such environments, the integrity of information exchanged between agents and system components becomes an important security consideration.

This paper examined a prototype approach that combines LSB steganography with HMAC-SHA256 authentication. The experimental implementation demonstrated that a message could be concealed within an image, subsequently extracted, and verified using cryptographic authentication. The controlled MITM simulation further demonstrated that unauthorized modification of an authenticated message results in authentication failure.

The findings support the usefulness of message authentication as an additional security layer in agentic AI communication. At the same time, the research demonstrates that authentication is not a complete solution. It cannot independently determine whether an original message is malicious, whether a sender is authorized to execute a particular operation, or whether a message is being replayed.

Therefore, secure agentic AI requires a layered approach. Message integrity should be combined with secure key management, least-privilege access, authorization, prompt-injection detection, replay protection, monitoring, adversarial testing and human oversight for high-risk activities. This layered model is consistent with the broader direction of contemporary AI security guidance.

The principal contribution of the paper is consequently practical rather than algorithmic: it demonstrates how established information-hiding and cryptographic techniques can be integrated into an agentic communication workflow and used to make tampering visible before altered information is trusted by a receiving component. With further development and testing, the prototype can provide a foundation for more comprehensive security mechanisms for distributed and autonomous AI systems.

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